

A background on Pardus

Pardus, as a GNU/Linux distro, is known to be user-friendly, sleek, simple and spicy with lots of custom configuration tools written by the Pardus team. The distro is not based on any of the traditional base distros like Debian or Red Hat. It's completely written from scratch. The package management tool is PISI. The distro makes use of lzma compression for packages and also supports xdelta-like techniques to make fetching packages for installation, faster. Pardus has developed its own core libraries and APIs to develop different applications on the Pardus platform.

Personally, I have been fascinated by Pardus for the last two years. Its boot speed is what I found to be the most impressive. This had prompted me to dig into the `initrd/init` boot scripts. I found it interesting that all system configs and automation scripts (traditionally written in Bash) have been replaced by Python. Even before Ubuntu's fastboot technology Upstart came out, Pardus' init was much faster.

What's my project about?

We have numerous variants of GNU/Linux (popularly known as distributions or distros) for different purposes. When we go for a distro install, we receive a base system with a standard set of applications bundled along with them. To make the OS work to our likes and specs, we need to install another set of packages that cater to our working domain or interest. Also, it requires setting up themes, wallpapers, copying our own files to the home directory, etc.

My project aimed at bringing out a Web-based distro 'cooker'. Users should be able to create their own custom distro builds by providing numerous custom options ranging from wallpapers to package selections. By locating the URL of this distro cooker (let's call it Pardusman), we could do the following customisation, using a simple Web interface:

- Home Page**
 - Sign up for a user account
 - Sign in to Pardusman
- Distro type**
 - Select the distro type: Live CD or Install CD? If it's Live, specify the user name, password, and host name
 - Provide a build project name
- Repository:** Select the package repository from the Pardus servers, to be used for the build.
- Languages:** Select language support to be included, besides setting the default language.
- Upload**
 - Upload a *RELEASE* file (i.e., a text file that appears in the *root* of the CD-ROM that contains some notes on the build).
 - Upload the contents of the home directory.
- Packages:** A package tree widget will appear. Packages



Figure 1: Sign-in



Figure 2: Distro type

- are classified as different components according to their group. For example, the GIMP and Inkscape are other graphic tools included in the package group 'multimedia-graphics'. By using a drop-down tree element, we access the packages under a package group and select the check boxes in order to include them. We can even include all packages in a package group by checking the group itself. Also, there is a search option available to look for packages. All the packages from the selected Pardus repository will be listed in the packages widget. There is also an option to calculate size—on clicking the Calculate button, you can get the total size for the build.
- Wallpaper:** You can either select the default option from the list of wallpapers, or upload your own.
- Media Selection:** You can select the required image output. For example, an ISO image, or virtualisation images like VMWare, Qemu, Virtualbox, etc. Once you have completed the wizard, you can submit